

Information Security

Unit- 6 Malicious Software and Intrusion

Malicious Logic :

- It is a software program designed to damage or do other unwanted actions on a computer system. It is a set of instructions that cause a sites security policy to be violated. These are the most sophisticated types of threats to computer systems that exploit vulnerabilities.
- Malicious software can be divided into two categories:

1. Independents :

- These are self contained programs that can be scheduled and ran by the operating system. Eg, Worms, Zombie etc.

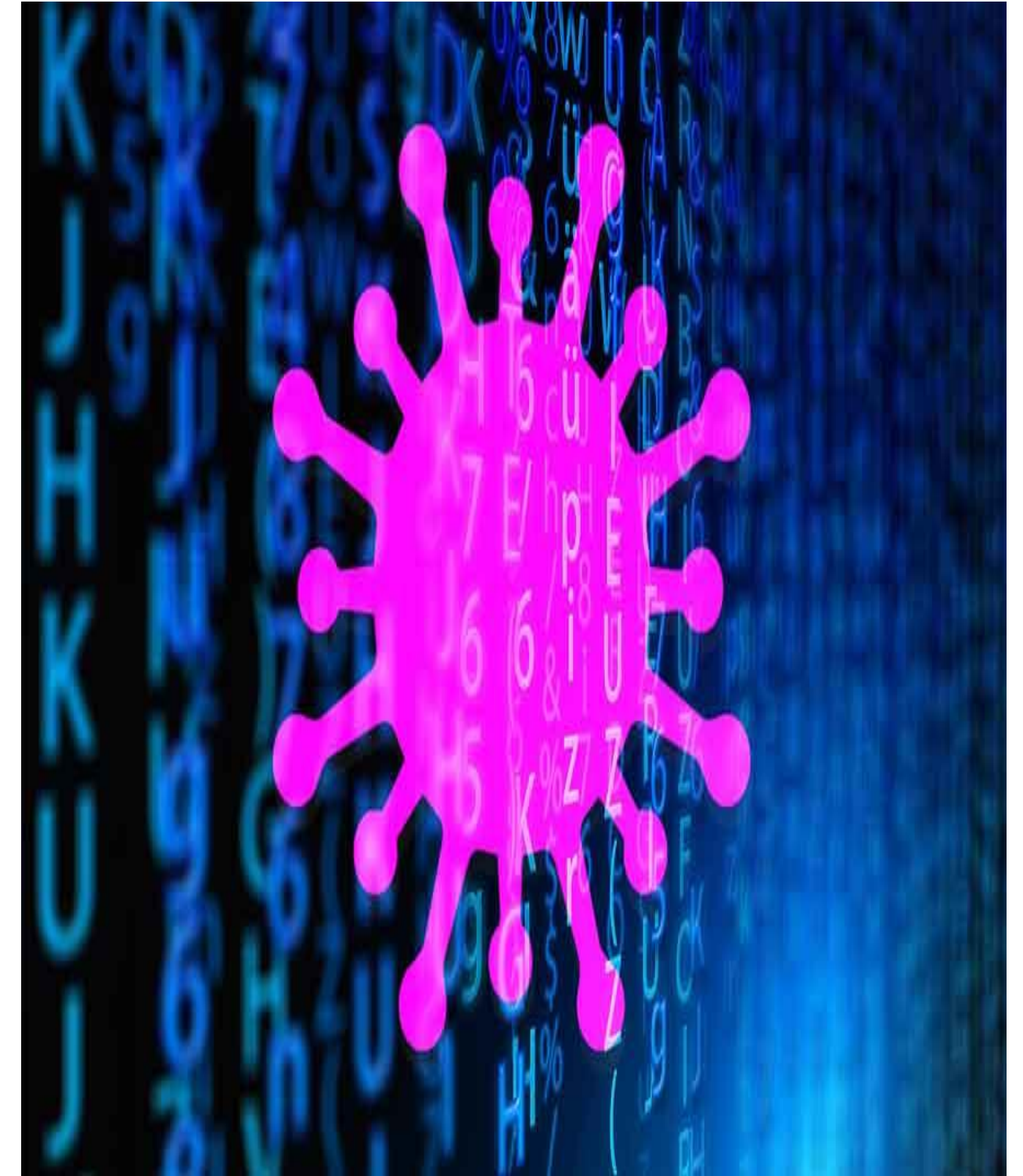
2. Needs Host Program :

- These are essentially fragments of programs that cannot exist independently of some actual application program, utility or system program. Eg, viruses, logic bombs etc.

Types of Malicious Logic

1. Virus:

- It is a type of malware which attach itself and modify the program and software.
- When it is executed, it replicates itself by modifying other computer programs and inserting its own code.
- When this replication succeeds, the affected areas are then said to be “infected” with a computer virus.
- It can damage hardware, software or files.



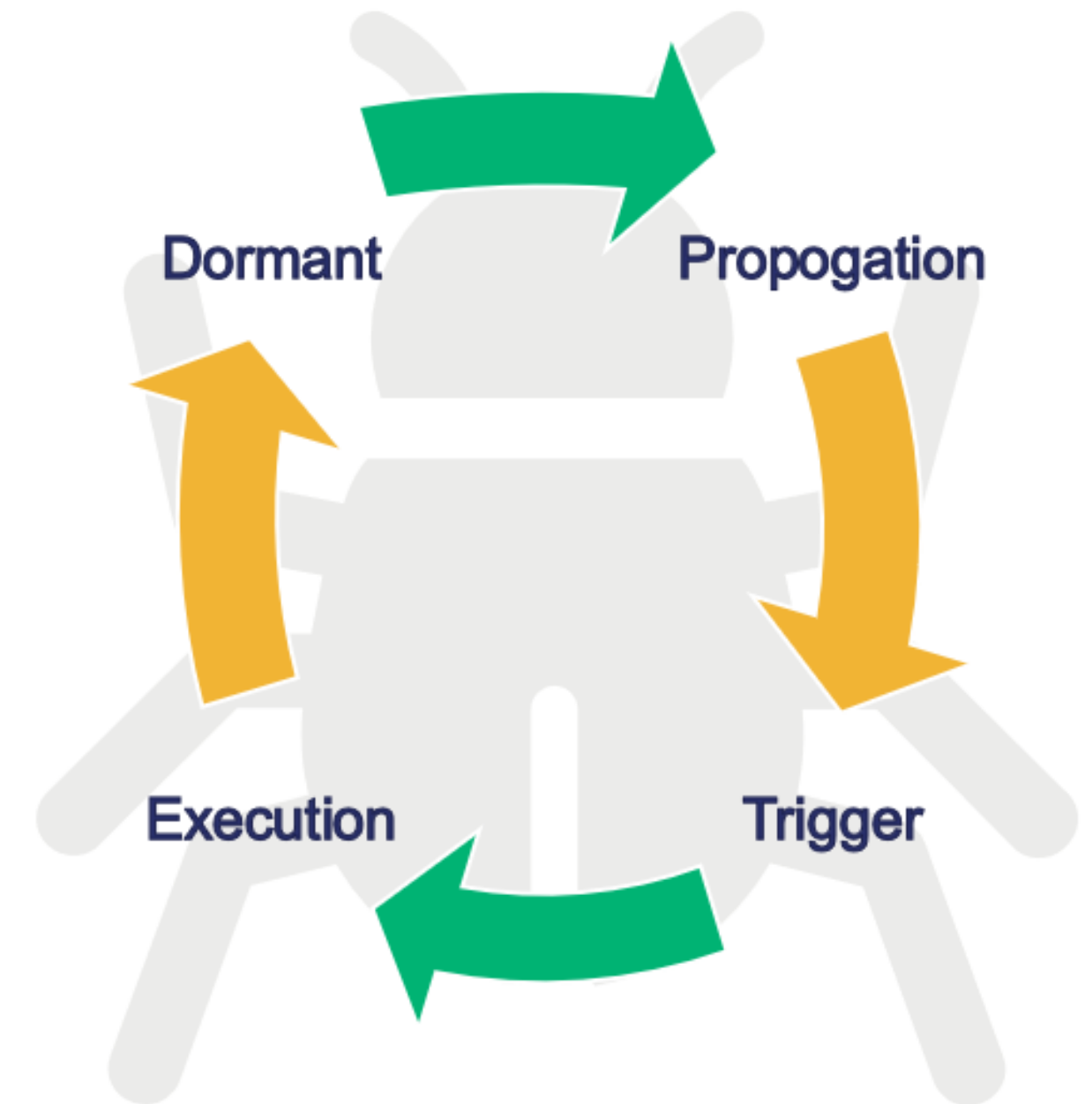
History of Virus

- In 1983-84 Fred Cohen from the university of southern California Wrote his paper “Computer Viruses-Theory and Experiments”. It was the first paper to explicitly call a self-reproducing program a “virus”, a term introduced by Cohen’s mentor Leonard Adleman.
- Vital Information Resources Under Seize.
- Creeper was the first programs that resembled an Internet worm. It ran on the old Tenex operating system and moved through the ARPANET(Advance Research Project Agency Network) . Creeper caused infected systems to print the message “ I’M THE CREEPER : CATCH ME IF YOU CAN.”
- Elk Cloner is the first personal computer virus or self-replicating program known to have spread in the wild on a large scale. In 1982, 15-year old Richard Skrenta wrote the virus for the Apple II operating system.

Phases of Virus

- When a virus inserts itself in a user's device, it stays dormant. It needs a trigger (i.e., a manual action performed by the host) to activate. Usually, the triggers can be activating the software, opening the infected files/media, enabling the macro, etc.
- Once active, the virus starts to search for other locations to replicate itself, such as on software, hard drives and removable hardware like memory cards and USB drives. It changes the original settings and codes of computer programs. Finally, it releases its payload. The payload is the core code that enables the virus to do what it was created to do.

Phases of the Computer Virus Lifecycle



Types of Virus

1. Parasitic Virus :

- It attach itself to executable files and replicates when the infected program is executed.

2. Memory Resident Virus :

- It resides in the main memory and infects every program that executes.

3. Boot Sector Virus :

- Infects a boot record and spreads when the system is booted from the disk containing virus .

4. Stealth Virus :

- A virus explicitly designed to hide itself from antivirus software.

5. Polymorphic Virus :

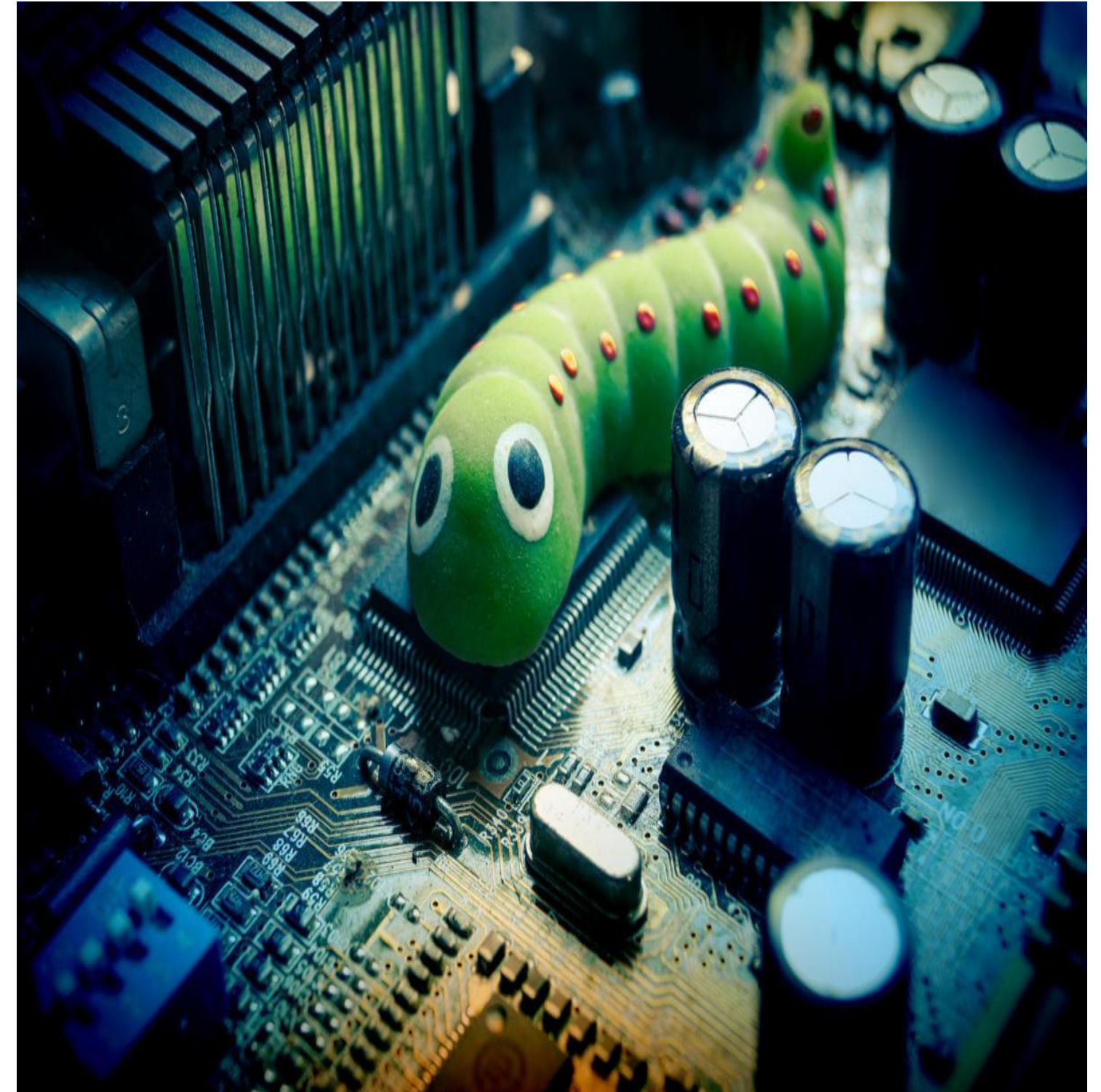
- A virus that mutates with every infection, making detection very difficult.

6. Metamorphic Virus :

- Mutates with every infection, rewriting itself completely at each iteration changing behavior or appearance, increasing the difficulty of detection.

2. Worms

- A computer worm is a self-replicating program that copies itself from one computer to another.
- It uses a computer network to send copies of itself to other nodes and do so without any user intervention. It searches for servers with security holes and copies itself.
- Unlike a virus, which needs to attach itself to a program, a worm is a standalone program that actively seeks out other machines to infect. Once it infects a machine, it uses that machine to continue spreading to more systems.
- Types of worm : Email worm, Internet worm, Instant messaging worm, File sharing network worm etc.



How Do Worms Spread?

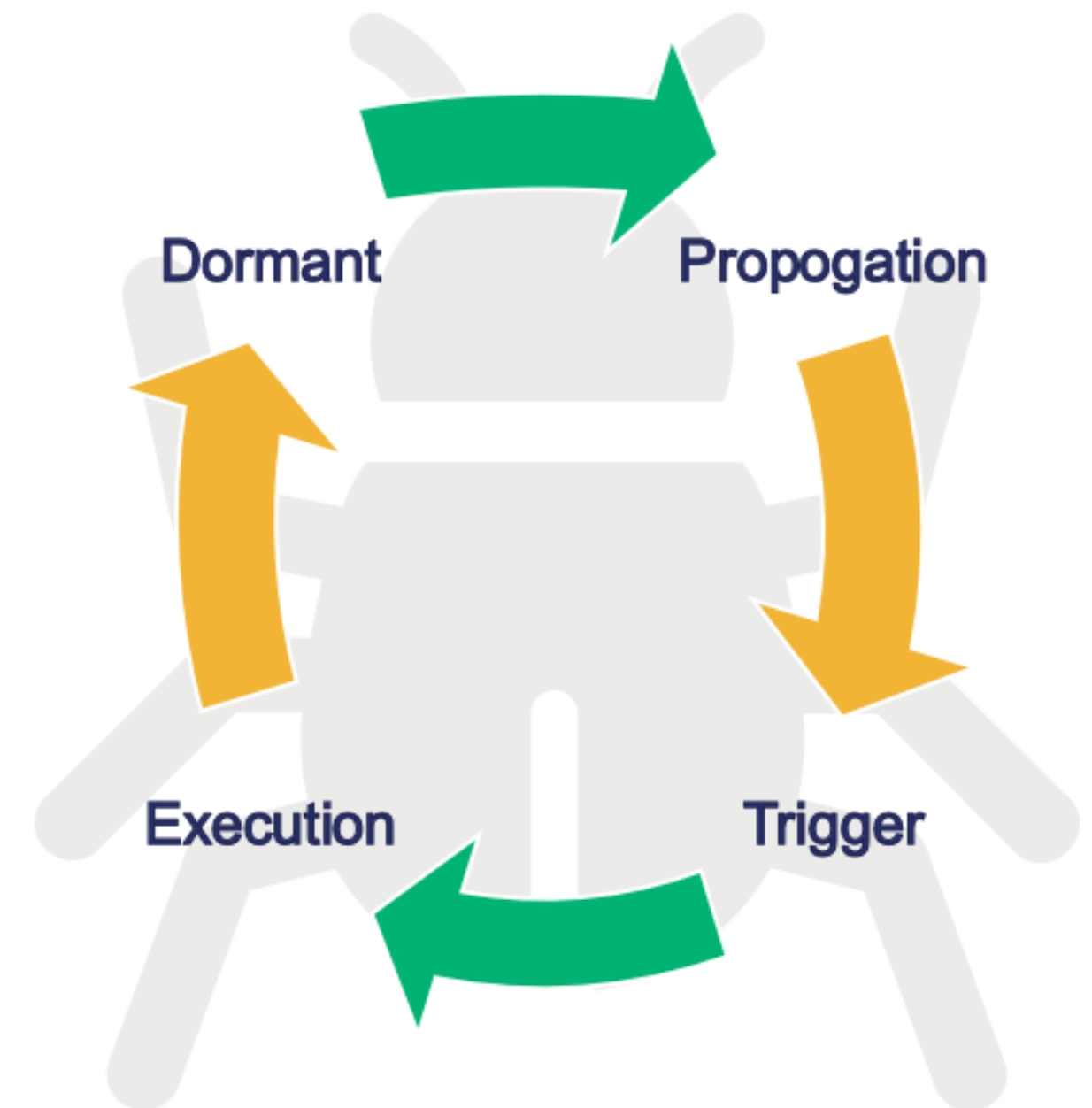
- Worms exploit vulnerabilities in software to gain access to new systems. They can spread in various ways, including:
 1. **Email and Instant Messages:** A worm can email itself to contacts or send itself through instant messages. When the recipient opens the email or message, the worm activates and spreads.
 2. **File Sharing:** Worms can infect files on removable media like USB drives. When these drives are connected to another system, the worm executes and spreads.
 3. **Remote Execution:** Worms can execute themselves on remote systems by exploiting vulnerabilities in network services.
 4. **Remote File Transfer:** Worms copy themselves to other systems via file transfer services, where they wait for a user to execute them.
 5. **Remote Login:** Worms can log in to remote systems using stolen credentials and then copy themselves to those systems.

Phases of a Worm's Life Cycle?

➤ A worm typically goes through the following phases:

1. **Dormant:** The worm is inactive.
2. **Propagation:** The worm spreads to other systems.
3. **Triggering:** The worm activates its payload.
4. **Execution:** The worm executes its payload, which could be anything from stealing data to causing damage.

Phases of the Computer Virus Lifecycle



Propagation Model for Worm

The propagation model of worms can be understood through a comparison with biological viruses. A simplified epidemic model can explain how worms spread:

$$\frac{dI(t)}{dt} = bI(t)S(t)$$

Where:

- $I(t)$ is the number of infected systems at time t .
- $S(t)$ is the number of susceptible (yet uninfected) systems at time t .
- b is the infection rate.

Phases of Worm Propagation

- **Slow Start Phase:** Initially, the worm spreads slowly because it starts from a single or a few hosts. The infection grows exponentially as each infected host infects more hosts.
- **Fast Spread Phase:** The number of infected hosts increases rapidly. Each infected host continues to infect more systems, leading to a high rate of infection.
- **Slow Finish Phase:** Most of the susceptible systems are already infected. The worm now has difficulty finding new targets, slowing the infection rate.

Types of Scanning Strategies for Worms

➤ Worms use different strategies to find new targets:

1. **Random Scanning:** The worm probes random IP addresses. This generates a lot of traffic and can be noticeable.
2. **Hit-List Scanning:** The worm uses a precompiled list of potential targets. This list allows the worm to infect systems quickly without drawing much attention.
3. **Topological Scanning:** The worm uses information from the infected host (like address books or contact lists) to find new targets.
4. **Local Subnet Scanning:** If the worm infects a system behind a firewall, it scans the local network for more targets, exploiting the local subnet structure.

3. Trojan Horse

- A Trojan horse is a type of malware that hides within a legitimate file or program to gain access to your device. Because Trojan malware is delivered inside a legitimate app or file, it's very difficult to detect. Trojans are used to spy on victims, steal data, infect other programs, and inflict other harm.
- Trojans are typically sent by scammers or hackers who use social engineering tactics, like the ones used in phishing attacks. Trojan horse malware appears as a harmless or even helpful file, leading users to install the malware unwittingly on their computers or phones.



Types of Trojan Horse

1. **Remote Access Trojan:** A Trojan horse designed to provide the attacker with complete control of victim's system.
2. **Data Sending Trojan :** A Trojan horse that is designed to provide the attacker with sensitive data such as password.
3. **Destructive Trojan:** A type of Trojan horse designed to destroy and delete files.
4. **Proxy Trojan:** A type of Trojan horse designed to use the victim's computer as proxy server.
5. **FTP Trojan:** A type of Trojan horse designed to open port 21 (port for FTP) and lets the attacker connect to your computer using FTP.
6. **Security Software Disable Trojan:** A type of Trojan horse designed to stop or kill security programs such as antivirus program without the user knowing.
7. **DOS Trojan:** A type of attack on network that is designed to bring the network to its knees by flooding it with useless traffic.

4. Zombies

- The term “Zombie” denotes a computer or electronic device compromised by malware or malicious software. Whenever a computer gets affected by malicious software then that computer can be controlled by the attacker sitting at some different location and the owner won't know about this. These infected computers are termed to be ‘zombies’.



Virus Vs Worms Vs Trojan Horse

Virus	Worm	Trojan Horse
A computer virus is a program that inserts itself into one or more files and then performs some actions.	A computer worm is a self-replicating program that copies itself from one computer to another.	A Trojan horse is a program with an overt (known) look and a covert (unwanted) effect.
Virus replicates itself.	Worms are also replicates itself.	Trojan horse does not replicate itself.
It cannot be controlled remotely.	It can be controlled remotely.	It can also be controlled remotely.
Spreading rate of viruses are moderate.	Spreading rate of worms are faster than virus and Trojan horse.	Spreading rate of Trojan horse is slow in comparison of both virus and worms.
It is used to modify the information.	It is used to halt the CPU and memory.	It is used to steal the user's information.

Intrusion

- Intrusion is any set of actions that attempts to compromise the confidentiality, integrity or availability of a computer resource.
- It simply refers to the unauthorized access or breach of a computer system, network, or application. This can involve bypassing security mechanisms to gain access to sensitive data, disrupt operations, or exploit system vulnerabilities for malicious purposes. Intrusions can be carried out by attackers (often referred to as hackers or cybercriminals) who use various methods and tools to penetrate systems.

Intruders

- Intruder is a person who attempts to gain unauthorized access to a system, to damage that system, or to disturb data on that system.
- The various types of Intruder are as follows-
 1. **Masquerader:** Means, a person who is not authorized to use computer but penetrates system protection by the way of legitimate user account.
 2. **Misfeasor :** A legitimate user who can access assets that he is not authorized to.
 3. **Clandestine user:** A user who gains supervisory access to the system, can be either inside or outside.
- The masquerader is likely to be an outsider; the misfeasor generally is an insider; and the clandestine user can be either an outsider or an insider.

Intrusion Detection System (IDS)

- Intrusion detection is the process of identifying and responding to malicious activity targeted at resource.
- It is a system that monitors network traffic for suspicious activity and issues alerts when such activity is discovered. It is a software application that scans a network or a system for harmful activity or policy breaching.
- IDS uses collected information and pre-defined knowledge-based system to reason about the possibility of an intrusion. It also provides services to cope with intrusion such as giving alarms, activating programs to try to deal with intrusion, etc.

Intrusion Analysis Approaches

- Intrusion Detection Systems (IDS) use two main approaches to analyze data and detect intrusions: Anomaly Detection and Signature or Heuristic Detection.

1. Anomaly Detection:

- Anomaly detection involves collecting data about the normal behavior of legitimate users over time. It then analyzes current behavior to see if it matches this normal behavior or if it is unusual (anomalous), potentially indicating an intruder.
- Working:
 1. **Training Phase:** A model of normal behavior is created by monitoring the system during normal operation.
 2. **Detection Phase:** Current behavior is compared to the model to classify it as either normal or anomalous.

Types of Anomaly Detection

1. **Statistical Methods:** Use statistical analysis of metrics (data points) to detect anomalies.
 - a. **Univariate Models:** Treat each metric separately.
 - b. **Multivariate Models:** Consider the relationships between different metrics.
 - c. **Time-Series Models:** Look at the sequence and timing of events.
2. **Knowledge-Based Methods:** Use expert-defined rules to classify behavior.
 - a. **Finite-State Machines:** Represent possible states and transitions.
 - b. **Description Languages:** Formalize the rules for classification.
3. **Machine Learning Methods:** Automatically create a model from training data.
 - a. **Bayesian Networks:** Represent probabilistic relationships.
 - b. **Markov Models:** Use states and transition probabilities.
 - c. **Neural Networks:** Simulate brain-like processing.
 - d. **Fuzzy Logic:** Handle uncertainty with approximate reasoning.
 - e. **Genetic Algorithms:** Use evolutionary techniques to create rules.
 - f. **Clustering and Outlier Detection:** Group similar data and identify outliers.

2. Signature or Heuristic Detection:

- This approach uses known patterns of malicious behavior (signatures) or predefined rules (heuristics) to identify intrusions. It is also known as misuse detection.

1. Signature-Based Detection:

- Matches data against a database of known malicious patterns.
- Commonly used in antivirus software and network intrusion detection systems (NIDS).

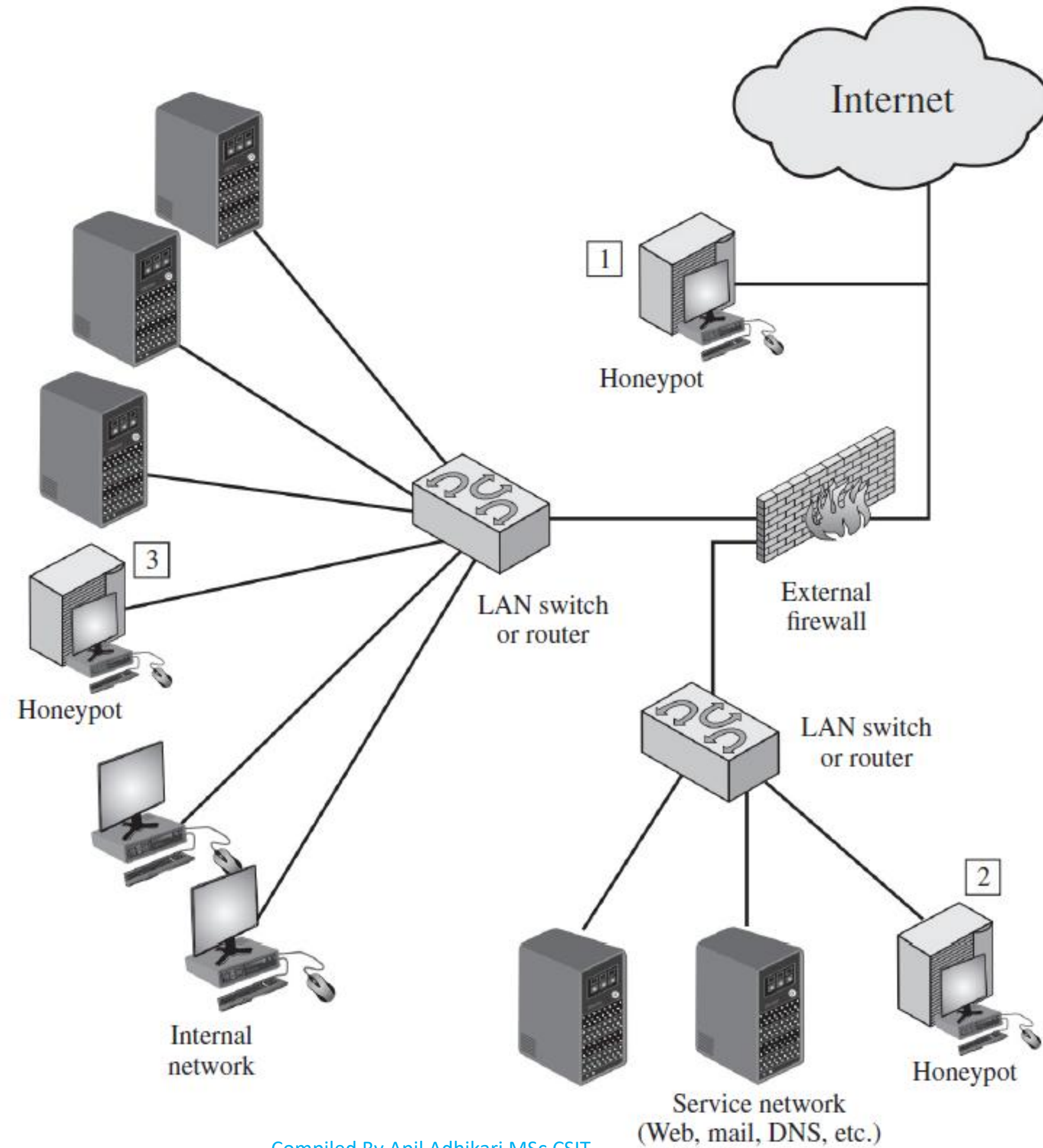
2. Heuristic-Based Detection:

- Uses rules to identify known attack patterns or suspicious behavior.
- Rules can be based on known vulnerabilities and attack methods.
- Security experts often develop these rules by analyzing attack tools and scripts.

Honeypots

- Honeypots are decoy systems designed to keep potential attackers away from critical systems, allowing security teams to monitor, log, and analyze the attacker's behavior. They are valuable tools in intrusion detection and can serve multiple purposes:
 1. **Divert Attackers:** Honeypots can distract attackers from valuable systems by presenting them with attractive but fake targets.
 2. **Collect Information:** They gather detailed data on the attacker's methods, tools, and actions.
 3. **Provide Time for Response:** Honeypots can keep the attacker engaged long enough for security teams to respond effectively.
- These decoy systems are designed to look like legitimate systems filled with fake data that seems valuable. Since no actual user should be accessing these systems, any interaction with a honeypot is immediately suspicious. Honeypots are equipped with tools that monitor and log the attacker's actions, providing useful information about the attack.

Honeypots



Types of Honeypots

- There are basically two types of honeypots:

1. Low Interaction Honeypots:

- These are simple systems that only pretend to offer certain services.
- They give attackers just enough to interact with but don't run full versions of those services.
- The main purpose is to catch attackers early in the process, before they reach real systems.

2. High Interaction Honeypots:

- These are real systems with actual operating systems and applications.
- They are designed to keep attackers busy for longer periods, allowing deeper observation.
- The main purpose is to analyze the attacker's methods and behavior.

Where Honeyypots Can Be Placed

- Honeyypots can be set up in different parts of a network, depending on what the organization wants to achieve and how much risk it's willing to take:
 1. Outside the External Firewall (Location 1)
 2. In the DMZ (Demilitarized Zone) (Location 2)
 3. Inside the Internal Network (Location 3)

!!!Thank you!!!